

Code-Layout, Readability and Reusability

Date	30 September 2026
Team ID	SWTID-2026-5807
Project Name	AI-StudyBuddy
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes / No / Partial)	Remarks
1	Consistent Indentation	Uniform spacing/tabs used throughout the code		4-space indent in Python (PEP 8), 2-space in React; auto-formatted with Black and Prettier
2	Proper File Structure	Files and folders are logically organized		Separate frontend and backend folders; code grouped by feature
3	Meaningful Variable Names	Variables reflect their purpose clearly		snake_case in Python and camelCase in JavaScript/React
4	Function / Method Names	Functions are descriptively named		Verb-based names such as generate_quiz() and fetchSummary()
5	Code Comments	Inline and block comments explain logic		Docstrings and short comments on key logic
6	Modular Design	Code is split into reusable functions/modules		Reusable services for auth, LLM calls and database access
7	No Redundant Code	Duplicate or unused code is removed		Shared helper functions; unused code removed using flake8 / ESLint
8	Error Handling	Exceptions and errors are handled gracefully		try/except blocks and clear API error responses

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level (High / Medium / Low)
1	LLM Client Wrapper	Python (FastAPI)	AI Tutor, Summarizer and Quiz Generator	High

2	Auth Module (JWT / OAuth2)	Python (FastAPI)	All protected API routes	High
3	Chat / Message UI Component	React.js	AI Tutor and Notes Summarizer pages	Medium
4	Database Utility (MongoDB)	Python (Motor / PyMongo)	Users, notes, quizzes, plans and progress	High
5	PDF / Text Extraction Helper	Python (PyPDF)	Notes Summarizer and Quiz Generator	Medium

Overall Code Quality Assessment:

Aspect	Rating (1-5)	Comments
Code Layout & Structure		Frontend and backend are kept in separate folders with clear feature-wise structure
Readability		Consistent naming and formatting enforced by Black / Prettier
Reusability		Shared services (auth, LLM wrapper, DB utils) reused across modules
Documentation / Comments		Docstrings and inline comments on important functions; README for setup
Overall Score		